

2.4 LABOUR SUPPLY AND PRODUCTIVITY

Multiple Choice Questions

1990/CE/II/20

Which of the following will increase the supply of labour?

- (1) The legal minimum working age is raised.
 - (2) The government introduces an unemployment benefit scheme.
 - (3) Workers are willing to work for longer hours at the same wage rate.
 - (4) The retirement age is raised.
- A. (1) and (2) only
 - B. (1) and (4) only
 - C. (2) and (3) only
 - D. (3) and (4) only

1991/CE/II/09

Which of the following will improve the average productivity of labour?

- A. a larger population
- B. longer working-hours
- C. more opportunities for tertiary education
- D. the emigration of skilled workers

1991/CE/II/56

The supply of labour does NOT depend on

- A. the size of the population.
- B. the composition of the population.
- C. the legal minimum working age.
- D. the productivity of workers.

1992/CE/II/10

Which of the following can increase the productivity of factory workers?

- (1) abandoning the piece-rate system
 - (2) installing more machinery
 - (3) importing labour from Mainland China
 - (4) providing a better working environment
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (4) only
 - D. (3) and (4) only

1992/CE/II/14

The Hong Kong government is expanding the tertiary education. This means that

- A. the provision of tertiary education no longer involves opportunity cost.
- B. the provision of tertiary education is not subject to diminishing returns.
- C. resources allocation will definitely be more efficient.
- D. the future labour productivity will increase.

1993/CE/II/06

Which of the following will improve the average productivity of labour in an economy?

- (1) extending compulsory education to a higher level
 - (2) raising the retirement age
 - (3) increasing the quantity of capital goods
 - (4) raising unemployment benefits
- A. (1) and (3) only
B. (1) and (4) only
C. (2) and (3) only
D. (2) and (4) only

1993/CE/II/19

The following table shows the output of dresses from a factory in 1991 and 1992.

Year	1991	1992
Number of workers employed	20	25
Units of dresses produced	90 000	80 000
Number of working hours per worker	3 000	2 000

The factory employed a _____ quantity of labour in 1992 and the average labour productivity was _____ in that year.

- A. larger higher
B. larger lower
C. smaller higher
D. smaller lower

1994/CE/II/17

Which of the following factors would tend to increase the labour supply in Hong Kong?

- (1) The government re-employs some retired civil servants.
 - (2) There are more public holidays each year.
 - (3) There is greater proportion of males in the population.
 - (4) The educational level of workers becomes higher.
- A. (1) and (3) only
B. (1) and (4) only
C. (2) and (3) only
D. (2) and (4) only

1995/CE/II/16

Which of the following will increase the average productivity of labour?

- (1) a larger labour force
 - (2) better education and training of workers
 - (3) better organization of production
 - (4) fewer public holidays
- A. (1) and (2) only
B. (1) and (4) only
C. (2) and (3) only
D. (3) and (4) only

1995/CE/II/59

Real GDP value-added per employee		
Year	1982	1992
Manufacturing sector	\$41 000	\$165 000
Services sector	\$103 000	\$280 000

Which of the following statements is **NOT** a conclusion drawn from the above table?

- A. The manufacturing sector had a higher labour productivity in 1992 than in 1982.
- B. The services sector had a higher labour productivity in 1992 than in 1982.
- C. The services had a higher labour productivity than the manufacturing sector in 1992.
- D. The services sector had a higher labour productivity growth rate than the manufacturing sector.

1996/CE/II/12

Which of the following will decrease the labour supply of a country?

- A. fewer job vacancies
- B. more universities place for students
- C. an increase in the retirement age of workers
- D. an increase in the number of immigrants

Refer to the table below for Question 13 and 14:

	Firm A	Firm B
Number of workers employed	200	120
Number of working hours per worker	1 000	1 900
Units of output produced	100 000	110 000

1996/CE/II/13

Which of the following comparisons is correct?

- A. Workers in Firm B has higher productivity.
- B. Firm B employed more labour services.
- C. Firm B enjoys a greater economies of scale.
- D. Production in Firm B is more capital-intensive.

1996/CE/II/14

Which of the following is a possible reason for the difference in labour productivity of a Firm A and Firm B?

- A. better training of workers in Firm A
- B. better working environment in Firm B
- C. a larger labour force in Firm A
- D. a higher wage rate in Firm B

1997/CE/II/55

An increase in the number of Form 7 graduates studying in the universities will result in

- (1) a decrease in the present labour supply.
 - (2) a decrease in the present unemployment rate.
 - (3) an increase in future average labour productivity.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

1998/CE/II/14

The average productivity of labour will increase if

- (1) there is an increase in the number of machines.
 - (2) there is an increase in the size of the labour force.
 - (3) there is an improvement in the health conditions of workers
 - (4) workers work more hours per day
- A. (1) and (2) only
B. (1) and (3) only
C. (2) and (4) only
D. (3) and (4) only

1998/CE/II/15

Which of the following would increase the labour supply of Hong Kong?

- A. The demand for imported workers increase
B. All women are not allowed to work overtime.
C. More employers provide child care services to their employees.
D. More job vacancies are available.

1999/CE/II/14

Which of the following would increase the labour supply of Hong Kong?

- A. The provision of more day-time child-care services
B. A change in the payment system from piece rate to time rate
C. An increase in the number of public holidays in a year
D. A decrease in the number of people migrating to other countries.

1999/CE/II/20

Refer to the following table:

	Firm A	Firm B
Number of workers employed	100	60
Number of working hours per worker	350	650
Output (Units)	140 000	156 000

Which of the following concerning Firm A and Firm B is correct?

- A. Firm B has a higher average labour productivity than Firm A.
B. Firm A has a higher average labour productivity than Firm B.
C. Firm A has large labour supply than Firm B.
D. Firm B has a large labour supply than Firm A.

2001/CE/II/13

Under which of the following situations would the labour supply in Hong Kong decrease?

- A. All universities in Hong Kong admit fewer full time students.
B. More Hong Kong companies provide training courses to their staff during office hours.
C. Fewer Hong Kong people are allowed to emigrate to the US.
D. The Hong Kong government provides subsidies to private firms for installing high-tech equipment.

2001/CE/II/15

Refer to the following table:

	Firm A	Firm B
Number of workers employed	60	40
Total number of working hours	1 200	1 000
Total output (Units)	10 800	9 500

Which of the following is a possible reason for the difference in the labour productivity of Firm A and Firm B?

- A. Firm B has a better working environment.
- B. Firm A has a higher degree of division of labour.
- C. Firm A enjoys greater economies of scale.
- D. Firm B uses a more labour intensive method of production.

2002/CE/II/14

A higher average labour productivity means a higher

- A. average output per worker.
- B. marginal product per man.
- C. total output per worker.
- D. total product per unit of labour.

2005/CE/II/16

Study the following table:

Economy	A	B	C	D
Number of workers	20	5	30	15
Weekly working hours per worker	5	25	5	20
Total output	11 000	12 000	9 000	10 000

Which economy has the largest labour supply?

- A. A
- B. B
- C. C
- D. D

2005/CE/II/17

Which of the following situations will increase the average labour productivity?

- A. employing more workers
- B. extending the working hours
- C. changing from time rates to piece rates
- D. reducing the degree of division of labour

2006/CE/II/14

Which of the following would raise the average labour productivity?

- A. a larger working population
- B. fewer public holidays
- C. a larger quantity of advanced equipment
- D. a lower degree of occupational mobility of labour

2007/CE/II/15

	Country A	Country B
Number of workers	100	X
Average working hours of workers per day	8	Y
Total output (unit)	7 200	7 200

In the above table, if the labour productivity in Country B is higher, the possible value of X and Y would be ____ and ____ respectively.

- A. 140 6
- B. 150 5
- C. 200 4
- D. 300 3

2009/CE/II/16

If the government raises the legal retirement age,

- A. the occupational mobility of workers will increase.
- B. the average labour productivity will increase.
- C. the supply of labour will increase.
- D. more division of labour will be practised.

2010/CE/II/18

Which of the following will increase the average labour productivity?

- (1) The number of promotion posts increases.
 - (2) The number of retired workers increases.
 - (3) Production becomes more capital-intensive.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

2010/CE/II/20

Which of the following would decrease the labour supply of Hong Kong?

- A. better and more job opportunities in mainland China
- B. the imposition of a minimum wage law for low-skilled workers in Hong Kong
- C. a lower average salaries tax rate in Hong Kong
- D. an increase in the retirement age in Hong Kong

2012/DSE/I/06

Which of the following policies will result in an increase in the labour force of Hong Kong?

- A. extension of free education from 9 years to 12 years
- B. an increase in the minimum capital requirement of the Capital Investment Entrant Scheme (資本投資者入境計劃)
- C. an increase in the immigration scheme quota for talented persons from the mainland
- D. introduction of maximum working hours

2012/DSE/I/10

Refer to the table below.

	Factory A	Factory B
Number of workers employed	60	10
Total number of working hours	2 000	5 000
Total output (units)	12 000	20 000

Which of the following is a possible reason for the difference between the labour productivity of Factory A and Factory B?

- A. Factory B provides more training for its workers.
- B. More machines are used in Factory B.
- C. The working environment in Factory A is poorer.
- D. Factory A pays its workers at piece rate while Factory B pays at time rate.

2014/DSE/I/09

Which of the following will increase the labour supply of an economy?

- A. More firms allow male employees to take paternity leave.
- B. More companies offer free transport to employees working overtime.
- C. Universities provide more full-time associate degree programmes for secondary school graduates.
- D. The government adopts maximum working hours for the civil servants.

2015/DSE/I/06

Refer to the following table:

	Factory A	Factory B
Number of full-time workers	25	0
Number of part-time workers	0	50
Average working hours of full-time workers	8	0
Average working hours of part-time workers	0	6
Total output (units)	12 000	15 000

- A. Factory A uses time rate while Factory B uses piece rate to pay its workers.
- B. Factory A provides more training for its workers.
- C. The part-time workers employed by Factory B are more skilled.
- D. Factory B uses more capital goods.

2016/DSE/I/08

The labour supply of a country will increase if

- A. more training is provided for the existing labour force.
- B. the legal minimum working age is lowered from 16 to 15.
- C. foreign direct investment increases.
- D. the government increases the salaries tax rate.

2017/DSE/I/11

The input-output relationship of a firm is as follows:

	Year 1	Year 2
Number of workers	30	35
Number of working hours per worker per day	10	8
Quantity of output (units)	3 000	2 900

Based on the above information, labour supply _____ and average labour productivity _____.

- A. rises rises
- B. rises falls
- C. falls rises
- D. falls falls

2018/DSE/I/09

The labour supply in Hong Kong would increase if

- A. more Hong Kong people migrated to foreign countries.
- B. more Hong Kong students studied in tertiary institutions.
- C. the Government lowered the qualification requirements for the Quality Migrant Admission Scheme.
- D. all working fathers were allowed to take two extra days of paternity leave.

2019/DSE/I/06

In the 2018-19 Budget, the Hong Kong Government raised the subsidy ceiling of the Continuing Education Fund from \$10 000 to \$20 000 per applicant to encourage members of the public to pursue further education.

This policy will lead to a rise in _____ and _____ in the future.

- A. labour supply.....geographical mobility
- B. labour supply.....occupational mobility
- C. labour productivity.....geographical mobility
- D. labour productivity.....occupational mobility

2023/DSE/I/06

Which of the following would raise both the labour supply and average labour productivity of an economy?

- A. The government raises the amount of unemployment benefits.
- B. Free education is extended to university programmes.
- C. More firms install advanced production equipment.
- D. The quota of an immigration scheme of highly skilled individuals is increased.

2024/DSE/I/08

The Hong Kong Government has implemented the ‘Top Talent Pass Scheme’ (高 端 人 才 通 行 證 計 劃) to attract foreign talents to work in Hong Kong. Successful applicants will be issued a two-year pass for exploring job opportunities in Hong Kong.

As a result of the implementation of the above scheme, _____.

- (1) Hong Kong’s labour supply will increase
 - (2) geographical mobility of foreign talents will increase
 - (3) average labour productivity of workers in Hong Kong will increase
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- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

2024/DSE/I/09

Refer to the following information about a toy-making firm.

	Year 1	Year 2
Number of workers (persons)	250	300
Number of working hours per worker per day (hours)	8	9
Total output per day (units)	9 000	10 800

Which of the following are possible reasons for the above change *in* the average labour productivity of the toy-making firm?

- (1) The wage payment method of the firm is changed from time rate to piece rate.
 - (2) The firm does not provide adequate training for new workers in Year 2.
 - (3) The working environment of the firm is improved in Year 2.
 - (4) The management of the firm becomes poorer in Year 2.
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- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (4) only
 - D. (3) and (4) only

2025/DSE/I/35

Suppose the nominal interest rate remains constant. When the actual inflation rate is lower than the expected inflation rate, the actual real interest rate will be higher than the_____.

- (1) expected real interest rate
 - (2) nominal interest rate
 - (3) actual inflation rate
-
- A. (1) only
 - B. (1) and (2) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

Short & Structured Questions

1990/CE/I/2(b)

The data below refer to Hong Kong's financial sector in 1985 and 1986:

	1985	1986
Value-added (in thousand dollars) at constant (1985) market prices	39 589 000	47 052 000
Employment (in thousands)	177	190

- (i) Use the above data to show how the average labour productivity in the financial sector changed from 1985 to 1986. (4 marks)
- (ii) Suggested **TWO** reasons for the change in (b)(i). (4 marks)

1994/CE/I/2

What is meant by the average labour productivity? Suggest **TWO** methods of raising labour productivity. (6 marks)

2000/CE/I/11(c)

Suppose Country B has a higher average labour productivity than Country A. Define average labour productivity and give **THREE** reasons for the higher average labour productivity of Country B. (5 marks)

2002/CE/I/3

	Firm A	Firm B
Number of workers employed	100	120
Number of working hours per worker	500	800
Output (units)	100 000	144 000

Calculate the average labour productivity of each firm and state **TWO** reasons why the two productivities are different. (4 marks)

2005/CE/I/10(a)(ii)

The two countries have a different output of watches, with each country using the same amount of resources in their production. State **THREE** possible reasons for such a difference. (3 marks)

2007/CE/I/2

Study the following information on a factory.

Year	2006	2007
Number of workers employed	40	50
Number of working hours per worker	300	200
Units of output produced	36 000	40 000

- (a) Calculate the labour productivity of the factory in 2006 and 2007, respectively. (2 marks)
- (b) Give **TWO** possible reasons to explain why the labour productivity has changed. (2 marks)

2010/CE/I/2

Refer to the following data of Firm A and Firm B. Both firms are engaged in the same industry.

	Firm A	Firm B
Number of workers employed	20	12
Number of working hours per worker	100	200
Number of output (units)	100 000	200 000

- (a) Explain whether Firm A has a larger labour supply than Firm B. (3 marks)
- (b) Explain whether Firm B has a higher average labour productivity than Firm A. (3 marks)

2024/DSE/I/13(d)

Hong Kong experienced an economic downturn because of the Covid-19 pandemic. It still faces various challenges on its path to resumption of normalcy after the pandemic.

Source A: Fiscal balance of the Hong Kong Government

Source B: Exchange rate of US dollar (USD) to Renminbi (RMB) before and after the pandemic

Date	Exchange Rate
1 April 2019	USD 1 = RMB 6.71
1 November 2023	USD 1 = RMB 7.32

Source C: News extract about the catering industry in Hong Kong in 2023

According to a representative of the catering industry in Hong Kong, local restaurants are unable to recruit enough waiters even though they are offering wages that are more than 50% above the statutory minimum.

Source D: Wage rates of waiters in Hong Kong in 2023

Statutory minimum wage rate	\$40 per hour
Average wage rate of waiters	\$62.4 per hour

Source E: Some data and views on the Hong Kong economy

 A banker	Influenced by the contractionary monetary policy in the US, the interest rate in Hong Kong rose drastically from mostly below 2% in 2018 to over 5% by the end of 2023.
 An economist	The inflation rate in the Mainland remained lower than that in Hong Kong throughout 2023. Negative growth in the general price level was recorded in October and November in the Mainland.
 A government official	Reviving the night-time economy is essential in facilitating the economic recovery in the short term, whereas innovation and technology development is a significant engine for driving high-quality economic growth in the medium-to-long term.

For part (d), candidates are required to present their answers in an essay form. Criteria for marking will include the use of the sources and economic theories, relevant content, logical presentation, and clarity of expression.

(d) Suppose there are two suggestions to stimulate Hong Kong's economic recovery and growth:

Suggestions	Objectives
I. Importing a large number of low-skilled workers for the catering industry and organising more night-time events to attract citizens and tourists to stay out at night for leisure	To facilitate the revival of the catering industry by boosting their revenue
II. Providing tax deductions for firms' expenses on innovation and technology projects to attract local and foreign investors to invest in the innovation and technology industry in Hong Kong	To enhance labour productivity and potential output

With reference to the above sources and your own knowledge in Economics, discuss

- how each of the above two suggestions may achieve its objective, and
- some possible factors that may limit the effectiveness of the two suggestions in achieving their corresponding objective. (Hint: You may consider some macroeconomic variables shown in the above sources, e.g. the exchange rate and the interest rate.)

(14 marks)